varying amounts of gravel and ranged from loose to very dense. The fill was characterized by its mottled texture and the presence of angular gravel.

**Pre-Olympia Deposits (Qpo)**: Underlying the fill, we encountered glacially consolidated soils we identified as consistent with Pre-Olympia Deposits. At the location of Boring B-1, the deposits consisted of five feet of dense to very dense poorly graded sand with silt overlying non-plastic, fractured very stiff silt. At the location of Boring B-2, we encountered medium dense to very dense silty sand with gravel and poorly graded sand with silt.

The subsurface information from our borings was used to develop a cross-section through the site which is provided as Figure 3, Generalized Subsurface Profile A-A'.

Our descriptions of subsurface conditions are based on the conditions encountered at the time of our exploration. Soil conditions between our exploration locations may vary from those encountered. The nature and extent of variations between our exploratory locations may not become evident until construction. If variations do appear, PanGEO should be requested to reevaluate the recommendations in this report and to modify or verify them in writing prior to proceeding with earthwork and construction.

### 3.4 Groundwater

Wet to water bearing soils were encountered in Boring B-2 at about 20 feet below grade while drilling. No groundwater seepage was encountered in Boring B-1, however we did encounter localized zones of increase soil moisture at 7½ and 12½ feet below grade. Based on the planned cuts of up to 15 feet deep, we do not anticipate groundwater will result in significant construction related issues. There will be fluctuations in seepage depending on the season, amount of rainfall, surface water runoff, and other factors. Generally, the water level is higher and seepage rates are greater in the wetter, winter months (typically October through May).

## 4.0 GEOTECHNICAL RECOMMENDATIONS

### 4.1 Seismic Design Parameters

The 2012 International Building Code (IBC) seismic design section provides a basis for seismic design of structures. Table 1 below provides seismic design parameters for the site that are in conformance with the 2012 IBC, which specifies a design earthquake having a 2% probability of occurrence in 50 years (return interval of 2,475 years), and the 2008 USGS seismic hazard maps.



Provided to Builders Exchange of WA, Inc. For usage Conditions Agreement see www.bxwa.com - Always Verify Scale